



Finals – Section 1 – Information Pack

Section 1: Make It

Relates to Questions 1 – 18

90 Marks available – Time available is 160 minutes

This section will be objectively assessed based on a combination of multi-choice questions, free-field response questions and the dynamic capability of your model. It will be marked out of 90.

It will be in your interest to pay attention to formatting as you go.

Everyone's raw mark for this section will be scaled up by an equal proportion such that the top scorer is awarded 100% of the 90 marks.

Section 2 requires you to use the model you build in Section 1. It is strongly recommended that you thoroughly read the Information Packs for both Section 1 and Section 2 before beginning construction of your Section 1 model.

INTRODUCTION

1. You work for a large manufacturer called ModelOff Manufacturing (MOM), which manufactures widgets. It manufactures eight different products through seven different factories. MOM has a single buyer who purchases all of its products and has a long term contract for the supply of them.
2. Your task for Section 1 is to develop a dynamic financial model to help optimize the future manufacture of the widgets (optimization is based on the marginal cost of production) and to assess the impact of machinery reaching the end of its life. The model is an annual model beginning on 1 January 2016 and ending on 31 December 2045 (30 years). The financial year of MOM matches the calendar year.
3. To assess this task, Section 1 is divided in two parts, Part A and Part B:
 - Part A covers questions 1 – 13 and assumes that no new machines are purchased to replace the machines that reach the end of their life.
 - Part B covers questions 14 – 18 and requires the inclusion of new machines in the model, as MOM are interested to know if they should be investing in replacement machinery when their current machinery reaches the end of its life.
2. In addition, the model you build for this Section 1 will be used in Section 2, where you will be required to ensure your model is well set out and well formatted. If you finish Section 1 early within the 160 minute allocation, you will be able to use any leftover time from Section 1 to make an early start on Section 2. For more information on Section 2, see the relevant Information Pack.
4. You will be required to submit **two** models for Section 1 at the end of 160 minutes; one that contains your answers and logic for Part A and a second that contains your answers and logic for Part B. You will also need to keep a local copy of your Section 1 models on your machine to enable you to do Section 2.
5. At the end of the allocated time for Section 1, there will be a 10 minute break where you will not be allowed to use your computer. You will then have 30 minutes to complete Section 2. If you complete all of Section 1 within the 160 minutes allocated, you may commence work on Section 2 early.
6. Question 18, does not require an answer. Instead, it allocates marks for a particular functionality in your model. To receive the marks for this question, your model will need to include a single cell as a dropdown 'switch' that, when changed, will correctly activate/deactivate logic for the introduction of the new machines required under Part B.



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7. All of the inputs discussed below have been provided to you in an Excel workbook “MO15 Finals - Sec 1 - Data - (your name).xlsx”.
8. The model will calculate the revenue and costs for each factory and each product. The model will not consider tax or financing and will not require the calculation of any financial statements.
9. All prices and expenses provided are expressed in 2015 dollars and are subject to annual indexation (see indexation section below).
10. The question breakdown is provided in the table below:

Section	Question	Marks
Part A - Expenses 10 marks	1	3
	2	3
	3	4
Part A - Capacity 9 marks	4	4
	5	5
Part A - Manufacturing Optimization 22 marks	6	4
	7	5
	8	7
	9	6
Part A - Overall 26 marks	10	6
	11	7
	12	5
	13	8
Part B 23 marks	14	3
	15	4
	16	5
	17	5
	18	6

NOTE:

- Part A – Expenses and Part A – Capacity should be able to be done independently of the other sections.
 - Subsequent sections will require you to have the previous section correct to proceed.
11. You should not round your calculations at any point in the model.
 12. The multiple choice answers presented to you may be rounded. Answer to the level of precision displayed in the answers (i.e. if your model calculation gave a result of 53.68292 and the options on the answer sheet are 53.6 or 53.7, you should select 53.7).
 13. To assist with formatting, a legend is provided in the ‘Cover’ sheet with most formats you will wish to use. A Custom Style for each format in the legend has also been added to the workbook.



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THE BRIEF – PART A

Manufacturing overview

14. MOM has seven factories identified by the districts they are located in: Telford, Ulsan, Victoria, Westminster, Xinzhou, York and Zwolle.
15. MOM manufactures eight different products. MOM refers to these products using the letters A, B, C, D, E, F, G and H.
16. Each widget is manufactured using an industrial machine, identified by the letter of the product it manufactures (i.e. Machine A produces Product A).
17. For legal reasons, no factory can have more than one of a particular type of machine at any point in time.
18. Each factory has a number of machines that are in use. A table has been provided showing the acquisition date of each machine (*Ass_General s2.01*). Assume that the machine was operational from the start of the year provided. Where no date is provided, that factory does not have a machine that manufactures that product.
19. Each machine has certain manufacturing characteristics: Initial Annual Rate, Useful Life, Final Year Efficiency, Electricity Usage and Labor Required (*Ass_General s2.02*). Electricity Usage and Labor Required are discussed under the Variable Expenses section below.
20. The number of products that each machine is capable of manufacturing per year degrades over time.
 - The Initial Annual Rate is the annual manufacture rate for the first year of a machine's operation (measured in thousands of products).
 - The Useful Life is the number of years the machine will function, after which it will be decommissioned. You do not need to model any costs or salvage value associated with the decommissioning.
 - The Final Year Efficiency is the machine's annual manufacture rate in the last year of operations, expressed as a percentage of the Initial Annual Rate.
 - A machine's annual manufacture rate between the first and final years of operations is calculated on a pro rata degradation basis (i.e. on a straight line basis) based on the first and final year annual manufacture rates.
 - For example, a machine with an Initial Annual Rate of 10 products, a useful life of five years and a Final Year Efficiency of 60% is capable of producing the following:
 - i. Year one, 10 products.
 - ii. Year two, nine products.
 - iii. Year three, eight products.
 - iv. Year four, seven products.
 - v. Year five, six products.
 - The degradation occurs in every year, regardless of the number of products the machine manufactures (including if the machine manufactures no products).
 - Do not round your calculations.



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Revenues

21. MOM has a long term contract with a single buyer who has contracted a specific volume of each product. This length of the contract is the same as the model length required.
22. The contract quantity required for each product is provided (*Ass_TimeSeries s1.01*).
23. The non-indexed price that applies for each product is provided (*Ass_General s2.03*). Prices are subject to indexation (see indexation section below).

Variable Expenses

24. Variable Expenses are directly related to the number of products manufactured. There are three variable expense categories:
 - **Materials:** Each product requires a combination of Plastic, Metal and Wood to manufacture. The volumes required are set in units of kg per product and vary by product (*Ass_General s2.04*). The price for Plastic, Metal and Wood is provided (*Ass_General s2.04*) and is subject to indexation (see indexation section below).
 - **Electricity:** There are two components to the electricity expense as follows:
 - a. Each machine requires a set number of Megawatt-Hours (MWh) per day to run (*Ass_General s2.02*). This does not change as the machine degrades over time. The price per MWh of electricity varies by factory and is subject to indexation (*Ass_General s2.05*). Assume there are 365 days in every forecast year.
 - b. All Metal used requires treatment before it can be used to manufacture products. This treatment requires 2 MWh of electricity per kg of Metal (*Ass_General s2.04*).
 - **Labor:** Each machine requires a set number of labor hours per day to run. (*Ass_General s2.02*). This does not change as the machine degrades over time. The price per hour of labor varies by factory and is subject to indexation (*Ass_General s2.05*). Assume there are 365 days in every forecast year.

Fixed Expenses

25. Fixed Expenses are incurred regardless of the volume produced. The only fixed expense is Overhead. The Overhead charge varies by factory and is subject to indexation (*Ass_General s2.05*).
26. The relevant Overhead is incurred in every factory in every period, even if the factory is not manufacturing any products.

Penalties

27. If the contracted quantity is not supplied, MOM will be required to make a penalty payment to the contract counterparty. This payment is equal to 10% of the total revenue that would have been paid (if they were supplied) for the products not supplied. MOM does not receive revenue for contracted volume not supplied.

Indexation

28. All prices and expenses provided are expressed in 2015 dollars. They are subject to annual indexation from the first forecast year (2016).
29. There are three different indexation assumptions that apply to the prices for the products:
 - Price Index 1 increases by 4% per annum from 1 January 2016, it applies to Product A and Product H.



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- Price Index 2 increases by 3.5% per annum from 1 January 2016, it applies to Product C, Product D and Product G.
- Price Index 3 increases by 2.5% per annum from 1 January 2016, it applies to Product B, Product E and Product F.

30. There are three different indexation assumptions that apply to the various expenses:

- Cost Index 1 increases by 2% per annum from 1 January 2016, it applies to the per kg cost of Plastic, Metal and Wood. It also applies to the per MWh cost of electricity (by factory).
- Cost Index 2 increases by 3% per annum from 1 January 2016, it applies to the per hour cost of labor (by factory).
- Cost Index 3 increases by 4.5% per annum from 1 January 2016, it applies to the Overhead charge.

31. The annual indexation rates are provided (*Ass_General s2.06*).

Optimization

32. In years where MOM has greater capacity to manufacture a given product than it is required to under its supply contract, it must decide which machines to run part time, or not at all, to reduce the output to the level required under the supply contract. The following steps are followed by MOM to determine this reduction:

- The optimization calculation occurs for each year in isolation of all other years.
- The total capacity to produce each product is calculated. Where the total capacity is less than the total volume required under the supply contract, all machines for that product are run at maximum capacity.
- Where the total capacity is more than the total volume required under the supply contract, MOM will operate machines so that it incurs the lowest marginal cost of production. Marginal Cost is calculated as the total variable costs per product (excluding overhead). This is calculated for each machine in each factory.
- Your model can assume that there will never be a situation where two machines producing the same product in the same year have an equal marginal cost.
- Machine operating hours are turned down beginning with the machine in the factory with the highest marginal cost until the level of output matches that required under the supply contract.
- A machine can operate for a fraction of a year where needed to produce the correct volumes for the supply contract. Do not round any calculations (partial units of all values are permitted).



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THE BRIEF – PART B

Replacement machines

33. For Part B, you need to include machines replacements. When any machine reaches the end of its life, it may be replaced by a new machines at the start of the following year. This replacement only occurs if, without the replacement, MOM would not be able to supply the required volumes of the product in the year of replacement.
34. If there is sufficient capacity without replacing the machine, it is put on hold. If a machine replacement is put on hold, the replacement can occur at any future year under the conditions described above. That is, if a replacement is on hold and if there is a year where MOM would not be able to supply the required volumes without replacement, the machine is replaced at the start of that year.
35. If there is more than one replacement machine of the same type on hold, and a replacement machine is required, the machine that has been on hold the longest is always replaced first.
36. You can assume that there is never a need to replace more than one machine of the same product type in any given year.
37. A new machine has the same characteristics as the machine it replaces.
38. You do not need to include any cost for the purchase or installation of the new machines.
39. The answers for Part B are **not** multi-choice. Please write your answer and a model reference in the space provided.
40. Remember that there are marks available in the final question for modeling your Part B such that the machine replacement logic can be included/excluded through a single drop-down switch cell.



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Your Name: _____

QUESTIONS (CIRCLE YOUR ANSWER)

PART A – EXPENSE QUESTIONS

Question 1 (3 Marks)

What is the cost per MWh of Electricity for the York factory in 2043?

- | | | |
|-----------|-----------|-----------|
| a) \$2.40 | d) \$2.43 | g) \$2.46 |
| b) \$2.41 | e) \$2.44 | h) \$2.47 |
| c) \$2.42 | f) \$2.45 | i) \$2.48 |

Question 2 (3 Marks)

What is the total cost of Plastic, Metal and Wood (excluding the electricity treatment of the Metal) required to manufacture one unit of Product A in 2019?

- | | | |
|------------|------------|------------|
| a) \$28.78 | d) \$28.81 | g) \$28.84 |
| b) \$28.79 | e) \$28.82 | h) \$28.85 |
| c) \$28.80 | f) \$28.83 | i) \$28.86 |

Question 3 (4 Marks)

What is the cost of the Metal treatment required to manufacture one unit of Product E at the Xinzhou factory in 2028?

- | | | |
|-----------|-----------|-----------|
| a) \$6.04 | d) \$6.07 | g) \$6.10 |
| b) \$6.05 | e) \$6.08 | h) \$6.11 |
| c) \$6.06 | f) \$6.09 | i) \$6.12 |



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PART A – CAPACITY QUESTIONS

Question 4 (4 Marks)

The Victoria factory acquired a Product G machine in 2008. What is the maximum number of Product G this machine can manufacture in the 2022 year?

- | | | |
|------------|------------|------------|
| a) 158,081 | d) 158,084 | g) 158,087 |
| b) 158,082 | e) 158,085 | h) 158,088 |
| c) 158,083 | f) 158,086 | i) 158,089 |

Question 5 (5 Marks)

What is the maximum number of Product C that MOM can manufacture across all factories in the 2018 year?

- | | | |
|------------|------------|------------|
| a) 510,826 | d) 510,829 | g) 510,832 |
| b) 510,827 | e) 510,830 | h) 510,833 |
| c) 510,828 | f) 510,831 | i) 510,834 |



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PART A – MANUFACTURING OPTIMIZATION QUESTIONS

Question 6 (4 Marks)

What is the Variable Cost required to manufacture one unit of Product C at the Ulsan factory in 2028?

- | | | |
|------------|------------|------------|
| a) \$68.86 | d) \$68.89 | g) \$68.92 |
| b) \$68.87 | e) \$68.90 | h) \$68.93 |
| c) \$68.88 | f) \$68.91 | i) \$68.94 |

Question 7 (5 Marks)

What is the Variable Cost required to manufacture one unit of Product H in 2018 at the factory which has the lowest marginal cost?

- | | | |
|------------|------------|------------|
| a) \$55.26 | d) \$55.29 | g) \$55.32 |
| b) \$55.27 | e) \$55.30 | h) \$55.33 |
| c) \$55.28 | f) \$55.31 | i) \$55.34 |

Question 8 (7 Marks)

How many factories will MOM operate to manufacture Product C in 2022 in order to meet the volumes under the supply contract?

- | | | |
|------|------|------|
| a) 0 | d) 3 | g) 6 |
| b) 1 | e) 4 | h) 7 |
| c) 2 | f) 5 | i) 8 |

Question 9 (6 Marks)

Which is the first Product where MOM is unable to meet the volumes required under the supply contract?

- | | | |
|--------------|--------------|------------------------------|
| a) Product A | d) Product D | g) Product G |
| b) Product B | e) Product E | h) Product H |
| c) Product C | f) Product F | i) Volumes can always be met |



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PART A – SUMMARY QUESTIONS

Question 10 (6 Marks)

Excluding profit penalties and excluding overheads, what is the total profit (in \$000s) made from selling Product B in 2041?

- | | | |
|-------------|-------------|-------------|
| a) \$42,733 | d) \$42,736 | g) \$42,739 |
| b) \$42,734 | e) \$42,737 | h) \$42,740 |
| c) \$42,735 | f) \$42,738 | i) \$42,741 |

Question 11 (7 Marks)

Excluding profit penalties but including overheads, what is the total profit (in \$000s) for the Zwolle factory in 2026?

- | | | |
|------------|------------|------------|
| a) \$9,492 | d) \$9,495 | g) \$9,498 |
| b) \$9,493 | e) \$9,496 | h) \$9,499 |
| c) \$9,494 | f) \$9,497 | i) \$9,500 |

Question 12 (5 Marks)

What is the total penalty payable (in \$000s) by MOM for volumes not delivered in 2040?

- | | | |
|--------------|--------------|--------------|
| a) \$129,285 | d) \$129,288 | g) \$129,291 |
| b) \$129,286 | e) \$129,289 | h) \$129,292 |
| c) \$129,287 | f) \$129,290 | i) \$129,293 |

Question 13 (8 Marks)

What is the total profit (in \$000s, including penalties and including overheads) for MOM over the 30 year analysis period?

- | | | |
|-----------------|-----------------|-----------------|
| a) \$11,767,460 | d) \$11,767,463 | g) \$11,767,466 |
| b) \$11,767,461 | e) \$11,767,464 | h) \$11,767,467 |
| c) \$11,767,462 | f) \$11,767,465 | i) \$11,767,468 |

You have reached the end of Part A. Save a copy of your model with logic for these answers activated before proceeding to Part B (you are required to submit a second model with your answers to Part B).



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PART B QUESTIONS

Question 14 (3 Marks)

What is the first year in which a Product B machine is replaced?

Answer: _____

Model reference (sheet and cell/s): _____

Question 15 (4 Marks)

What is the total number of Product G machines that get replaced over the 30 year analysis period?

Answer: _____

Model reference (sheet and cell/s): _____

Question 16 (5 Marks)

Measured to the nearest whole unit, what is the total capacity to manufacture Product F (including replacement machines) in 2029?

Answer: _____

Model reference (sheet and cell/s): _____

Question 17 (5 Marks)

Measured to the nearest thousand dollars, what is the total profit (including penalties and including overheads) for MOM over the 30 year analysis period?

Answer: _____

Model reference (sheet and cell/s): _____

Question 18 (6 Marks)

To receive the marks for this question, your model will need to include a single cell as a 'switch' that, when changed via a drop-down selection, will correctly add/remove the introduction of the new machines under Part B. The switch must enable the model to generate the correct answers to both Question 13 and Question 17.

Describe the location in your model of the switch and what value it should be set at to enable/disable machine replacement. Also describe the location of the answer to Question 13/Question 17.
